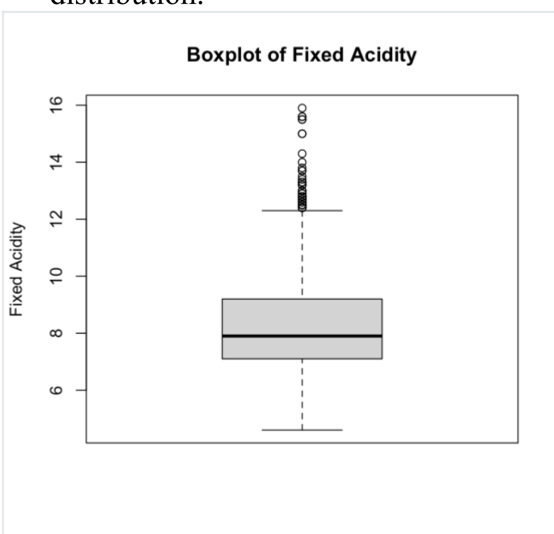
Red Wine Quality

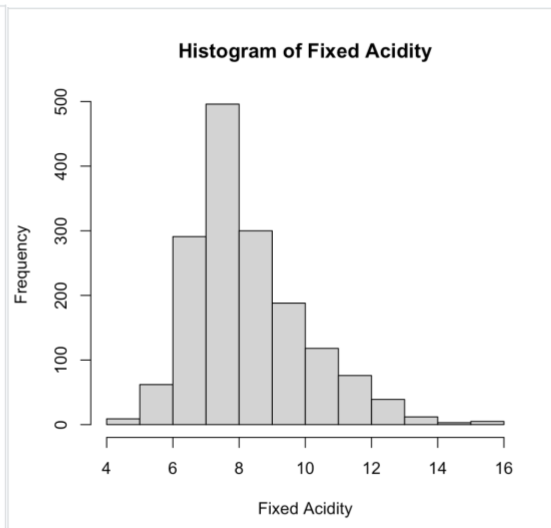
All the way from Portugal, *vinho verde* red wine was growing increasingly popular around the early 2000s. This dataset analyzes the physicochemical properties of this wine and how they might affect the quality of the wine when rated by experts. My goal is to analyze the data in order to find out what makes a *vinho verde* red wine "good."

Part One:

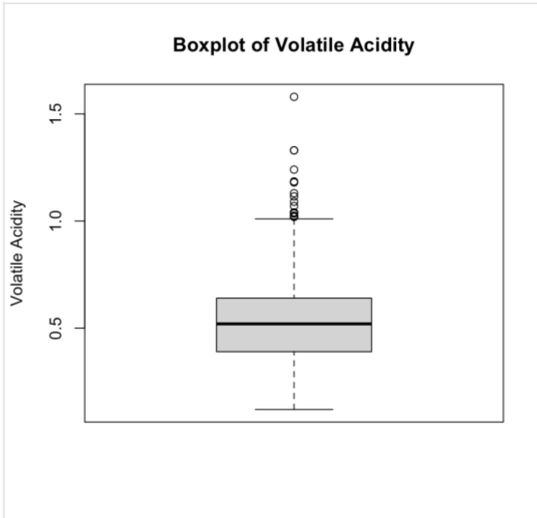
These are all the input variables based on physicochemical tests from the source. We can start by exploring the variables and defining them within context of the data using different graphs and summary statistics.

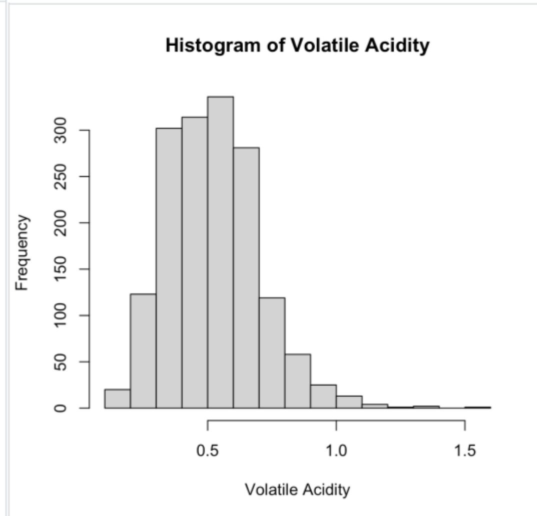
1. Fixed Acidity (g(tartaric acid)/liter)
   - This is most acids involved with wine, they are fixed or nonvolatile because they do not evaporate readily. (Cortez et al., 2009)
   - Min. 1st Qu. Median Mean 3rd Qu. Max.
   - 4.60 7.10 7.90 8.32 9.20 15.90
   - Based on the summary of this variable it is possible that there is a small outlier with a fixed acidity, but we can use a boxplot and histogram to better analyze the frequency distribution.





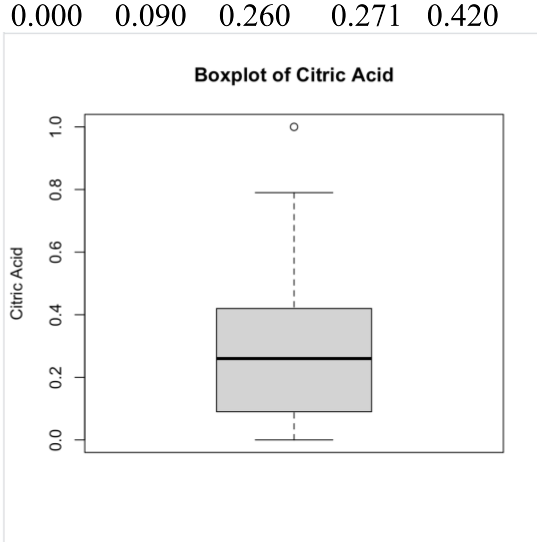
   - From these graphs we can see that the distribution of fixed acidity is right skewed.
   - From the summary statistics and graphs, we now know that many of the red wines tested had around 6-9 grams of tartaric acid per liter.
2. Volatile Acidity (g(acetic acid)/liter)
   - This is the amount of acetic acid in wine, which at too high of levels can lead to an unpleasant, vinegar taste. (Cortez et al., 2009)
   - Min. 1st Qu. Median Mean 3rd Qu. Max.
   - 0.1200 0.3900 0.5200 0.5278 0.6400 1.5800
   - Once again it seems that there is a right skewed distribution, but we can confirm with a boxplot and a histogram.

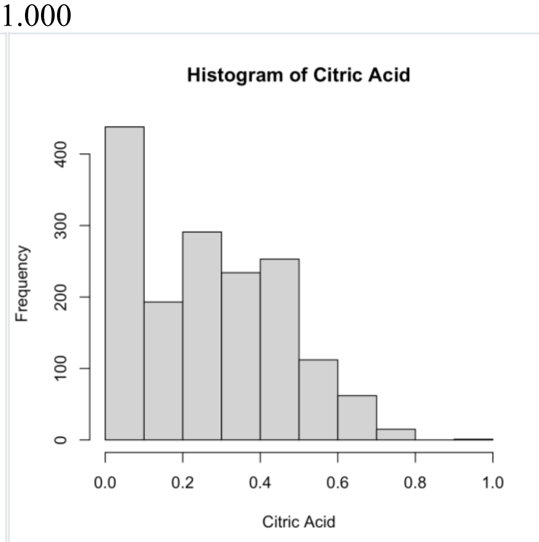




-
- From the mean and distribution, it seems that most of the red wines have about 0.2-0.7 grams of acetic acid per liter.

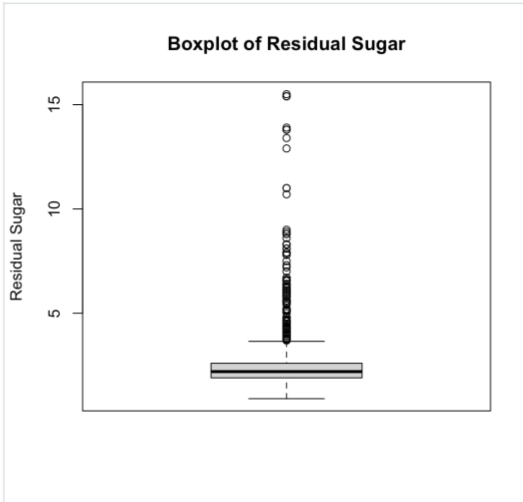
3. Citric Acid (g/liter)
   - Found in small quantities, citric acid can add 'freshness' and flavor to wines. (Cortez et al., 2009)
   - Min.   1st Qu. Median Mean   3rd Qu.   Max.
   - 0.000   0.090   0.260   0.271   0.420   1.000

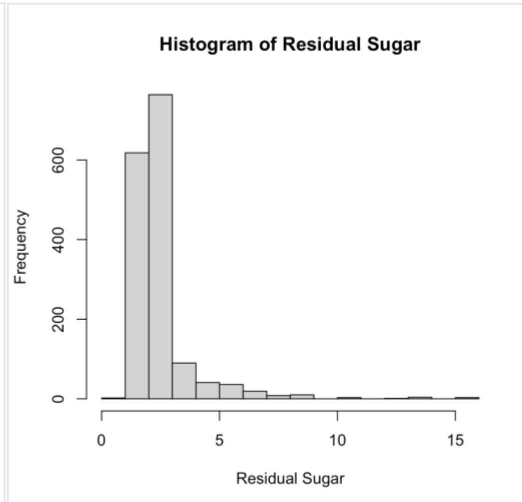




   -
   - Aside from a few outliers, it seems that a large majority of the red wine studied has about 0-0.4 grams per liter of citric acid.

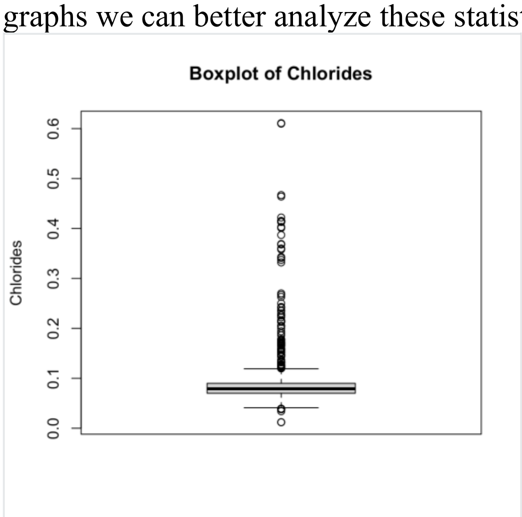
4. Residual Sugar (g/liter)
   - This is the amount of sugar remaining after fermentation stops, it's rare to find wines with less than 1 gram/liter and wines with greater than 45 grams/liter are considered sweet. (Cortez et al., 2009)
   - Min.   1st Qu. Median   Mean   3rd Qu.   Max.
   -   0.900   1.900   2.200   2.539   2.600   15.500

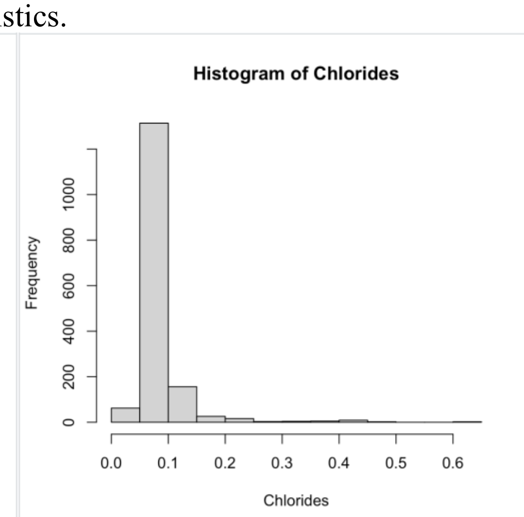




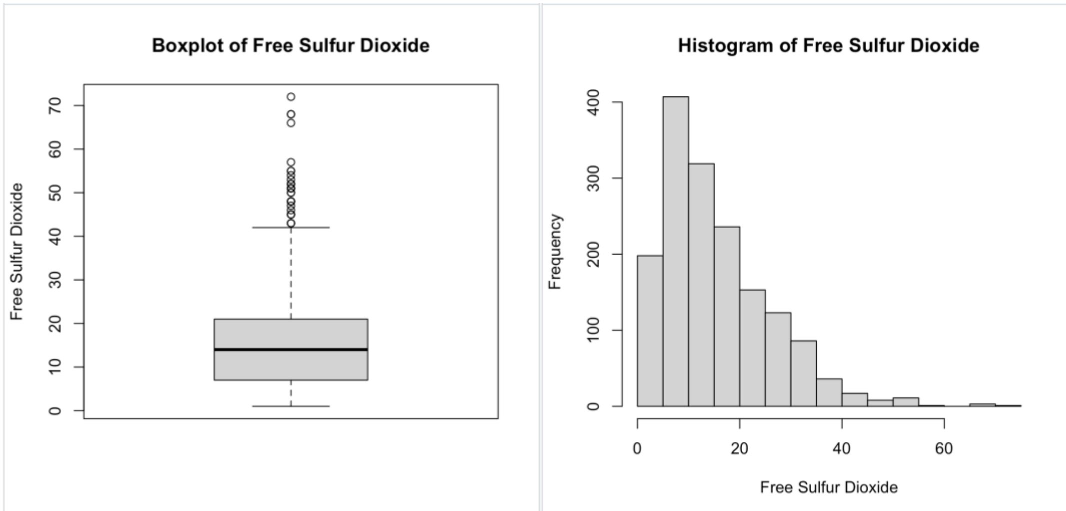
- With residual sugar we can see that larger outliers such as 15.5 g/liter and other values are skewing the statistics for this variable. Many of the wines have less than 5 grams/liter of residual sugar.

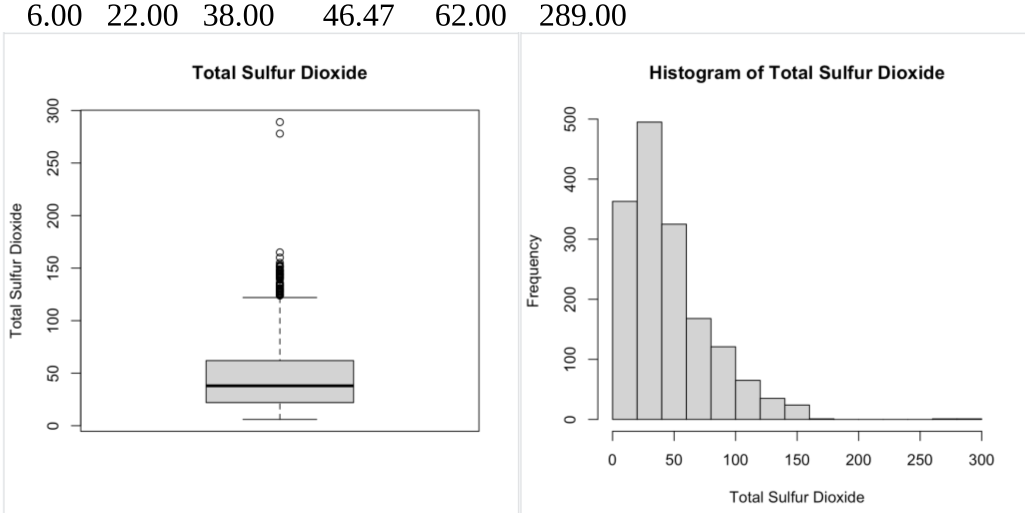
5. Chlorides (g(sodium chloride)/liter)
    - The amount of salt in the wine. (Cortez et al., 2009)
    - Min. 1st Qu. Median Mean 3rd Qu. Max.
    - 0.012 0.07 0.079 0.08747 0.09 0.611
    - There are very small amounts of sodium chloride per liter of red wine. Looking at graphs we can better analyze these statistics.



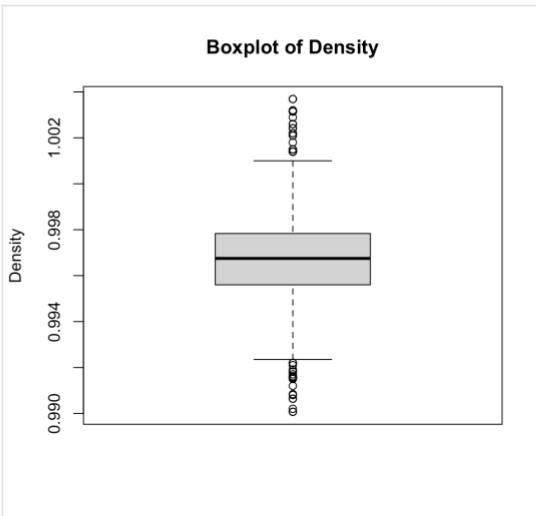


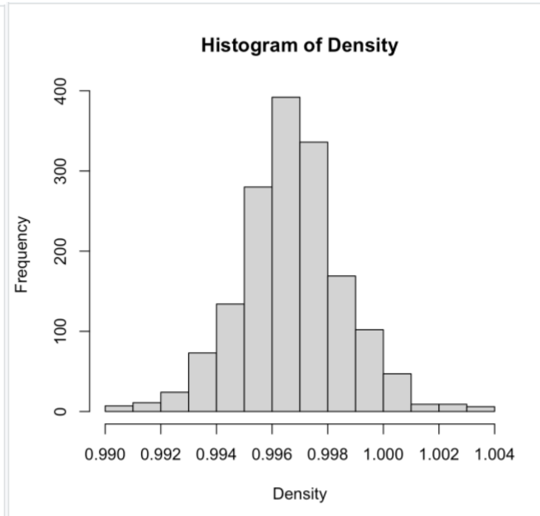
- Our assumptions based on the summary statistics were correct. Right around 0.9 grams/liter is the amount of salt most of these red wines have. The quartiles are very close which also suggests that this distribution curve is very narrow with a large right skew.

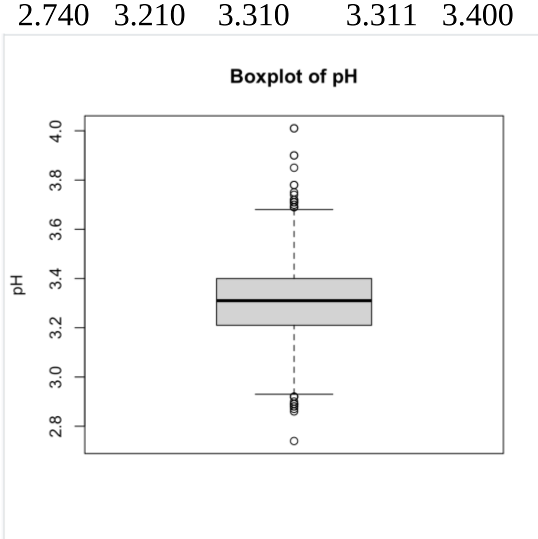
6. Free Sulfur Dioxide (mg/liter)
    - The free form of SO2 exists in equilibrium between molecular SO2 (as a dissolved gas) and bisulfite ion; it prevents microbial growth and the oxidation of wine. (Cortez et al., 2009)
    - Min. 1st Qu. Median Mean 3rd Qu. Max.
    - 1.00 7.00 14.00 15.87 21.00 72.00
    - 

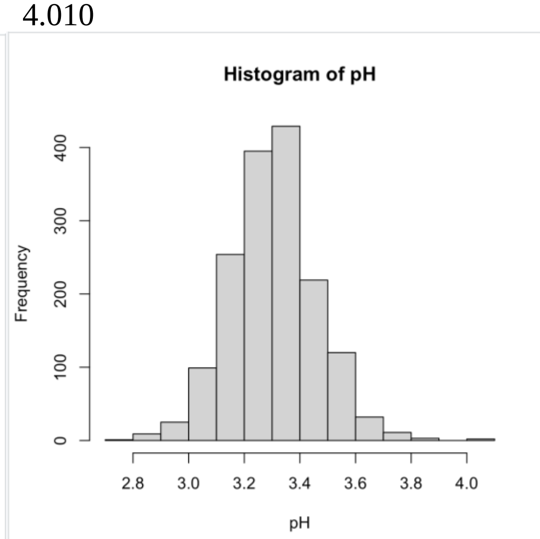
    - In another right skewed graph, we can see that most wines have around 13-18 mg/liter of free sulfur dioxide.
7. Total Sulfur Dioxide (mg/liter)
    - amount of free and bound forms of S02; in low concentrations, SO2 is mostly undetectable in wine, but at free SO2 concentrations over 50 ppm, SO2 becomes evident in the nose and taste of wine. (Cortez et al., 2009)
    - Min. 1st Qu. Median Mean 3rd Qu. Max.
    - 6.00 22.00 38.00 46.47 62.00 289.00
    - 

    - From the graph we can see most of the red wines will have below 100 mg/liter of total sulfur dioxide, with majority below 50 mg/liter which from the definition of sulfur dioxide in wine is unnoticeable. The large outlier may effect distribution using the central limit theorem.

8. Density (g/cubic centimeter)
   - the density of wine is close to that of water depending on the percent alcohol and sugar content. (Cortez et al., 2009)
   - Min. 1st Qu. Median Mean 3rd Qu. Max.
   - 0.9901 0.9956 0.9968 0.9967 0.9978 1.0037
   - 

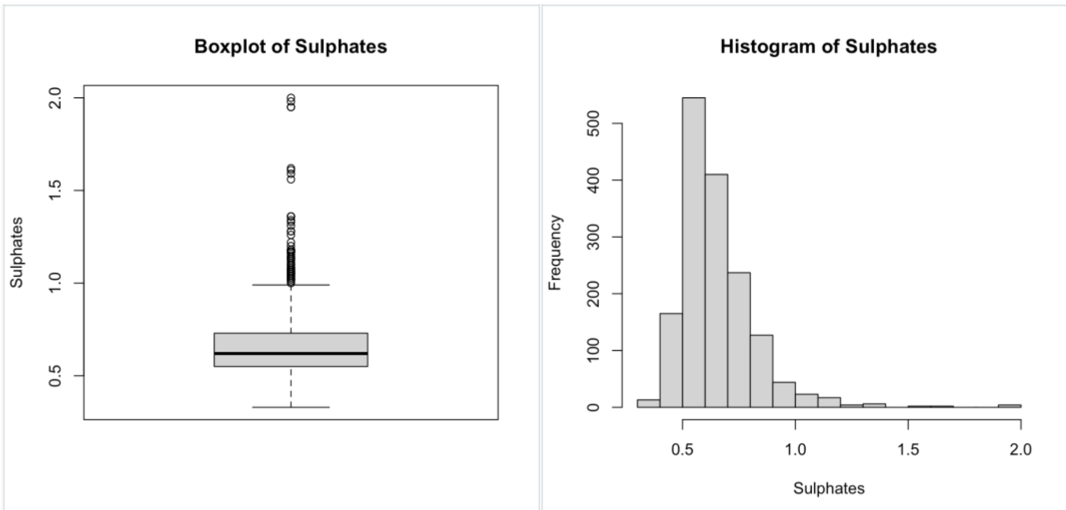


   - Based on the vary small amounts of variance in the summary statistics and the normal looking distribution we can say that the density of red wine is very similar in most red wines depending on the sugar and alcohol content.
9. pH
   - describes how acidic or basic a wine is on a scale from 0 (very acidic) to 14 (very basic); most wines are between 3-4 on the pH scale. (Cortez et al., 2009)
   - Min. 1st Qu. Median Mean 3rd Qu. Max.
   - 2.740 3.210 3.310 3.311 3.400 4.010
   - 

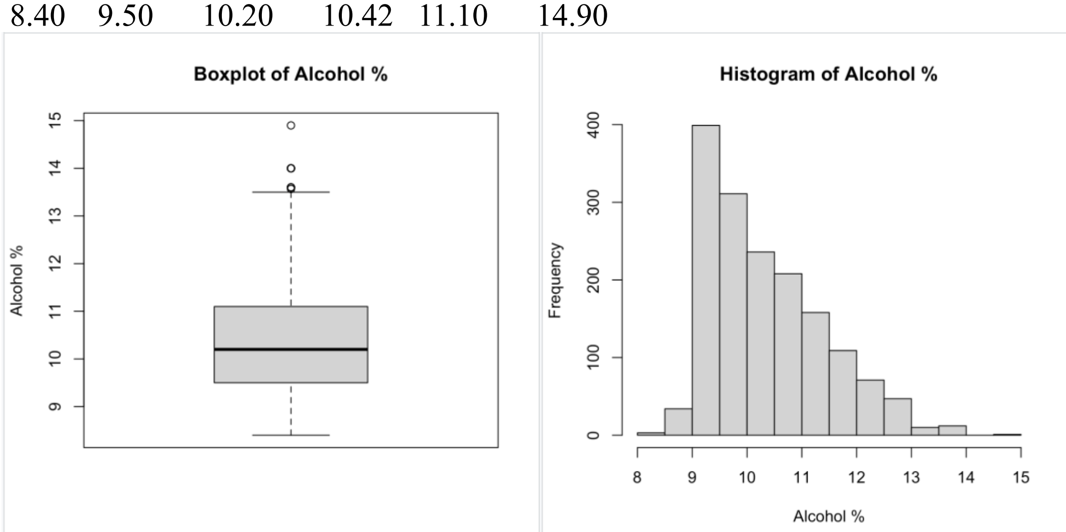


- Checking the assumption from the source's definition of the wines pH levels we can see that they are correct. The red wines they tested have pH levels normally distributed from 3-4.

10. Sulphates (g(potassium sulphate)/liter)
    - a wine additive which can contribute to sulfur dioxide gas (S02) levels, which acts as an antimicrobial and antioxidant. (Cortez et al., 2009)
    - Min. 1st Qu. Median Mean 3rd Qu. Max.
    - 0.3300 0.5500 0.6200 0.6581 0.7300 2.0000

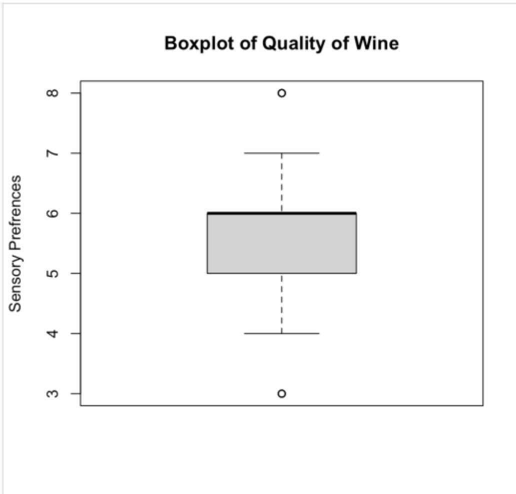


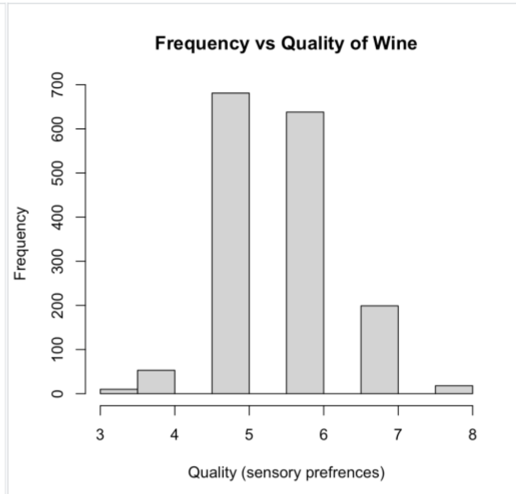
    -
    - Another right skewed graph shows us that most red wines tested have around 0.4-0.8 grams of potassium sulphate per liter.
11. Alcohol (vol.%)
    - the percent alcohol content of the wine. (Cortez et al., 2009)
    - Min. 1st Qu. Median Mean 3rd Qu. Max.
    - 8.40 9.50 10.20 10.42 11.10 14.90



    -

12. Quality (score between 0 and 10)
    - output variable (based on sensory data, score between 0 and 10) (Cortez et al., 2009)
    - Min. 1st Qu. Median Mean 3rd Qu. Max.
    - 3.000 5.000 6.000 5.636 6.000 8.000





    -
    - The quality of wine is measured as categorical data based on the sensory preferences of experts. We can see that most of the red wine tested was shown to be around a 5 or 6, but can we find a way to determine which input variable(s) created a wine of quality 8? That’s what we can determine in part two.

Part Two:

<u>Research Question #1.)</u>

After analyzing the many variables in part one, there are many points of interest in this dataset. To narrow down, many questions can be asked about the quality of the wine and what variables affect it the most. Indexing through the data and gathering two separate subsets. One with wine of quality less than or equal to five and one with wine of quality greater than five. Then comparing the alcohol percentage in "good" wine vs "bad" wine using a two-sample t-test. We can use the confidence interval and hypothesis test to come to conclusion whether the population means for alcohol percentages are the same in "good" wine or "bad" wine. This method for comparing the alcohol percentages is valid because a two-sample t-test compares the difference of averages between the two subsets. This difference can be used to find out how much alcohol percentage is used in a "good" wine.

$$H_0 = \mu_g = \mu_b$$

$$H_1 = \mu_g \neq \mu_b \; or \; \mu_g > \mu_b \; or \; \mu_g < \mu_b$$

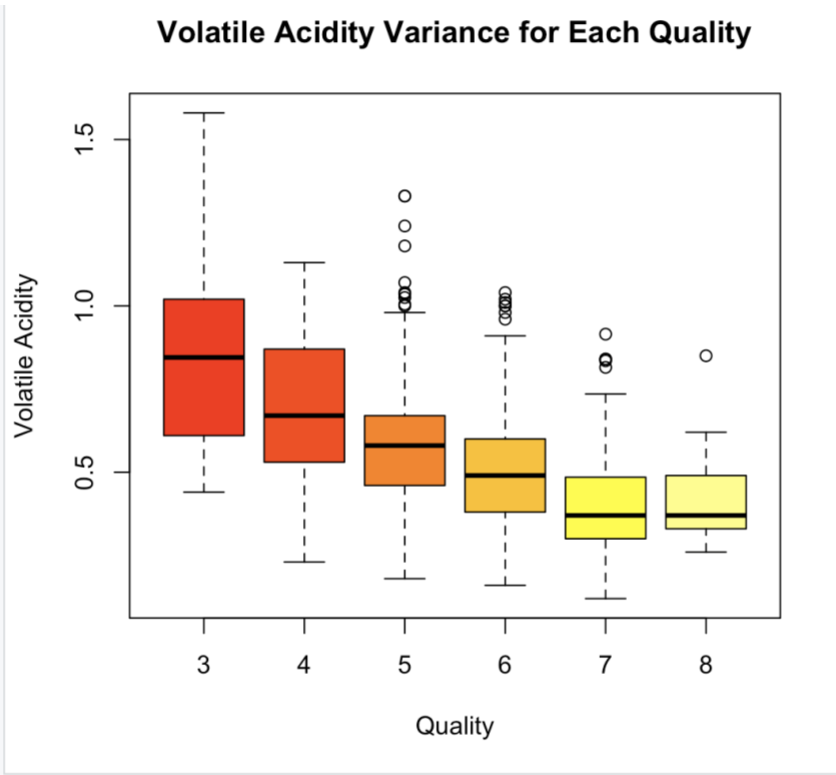
After preforming a t-test of the two different subsets, the code returns as the following.

```
> t.test(x = goodWine$alcohol, y = badWine$alcohol, alternative = "two.sided")

        Welch Two Sample t-test

data:  goodWine$alcohol and badWine$alcohol
t = 19.782, df = 1516.8, p-value < 2.2e-16
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 0.836479 1.020622
sample estimates:
mean of x mean of y
10.855029  9.926478
```

Analyzing the output, the p value is very low which means there is enough evidence to conclude that the mean percentage of alcohol in the population of "good" wine is different than the mean percentage of alcohol in the population of "bad" wine. Checking assumptions that are being made with this test, the sample size is large enough for the Central Limit Theorem kick in and there are no significant outliers. Further observing the values resulting from the t-test we are 95% confident that the alcohol percentage in "good" wine is higher than "bad" wine. This is because the confidence interval found is positive and does not include zero meaning we know that the difference of averages ("good" – "bad") will be more than zero 95 percent of the time when testing red wine samples.

Research Question #2.)

From the definition of volatile acidity or the presence of acetic acid, it is said that at too high level it leads to a vinegar taste in the wine. Do high levels of acetic acid in wine effect the quality of wine? To answer this question using the sample provided, we can analyze the variance of volatile acidity in the different qualities of wine. So, an ANOVA test seems like a valid way to approach this problem. First, a graph displaying boxplots of volatile acidity amongst the different qualities will be useful to see if performing and ANOVA will be insightful or not.



Looking at the plot it does seem like there could be a difference between the population means for volatile acidity of at least one quality of red wine. Next, we can perform the ANOVA to see if this is true or not.

```
> VolAnova = aov(vol_acid ~ as.factor(qual))
> summary(VolAnova)
                  Df Sum Sq Mean Sq F value Pr(>F)
as.factor(qual)    5   8.22   1.645   60.91 <2e-16 ***
Residuals       1593  43.01   0.027
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Since the F-value is a large enough value to show that the variance between the group variability is larger than the within group variability, the p-value is smaller than alpha of 0.05. This means that there is at least one population mean of volatile acidity that is different. In other words, all the population means aren't equal so we can reject that hypothesis. Knowing that one group has a different population mean we can use pairwise test with corrections of alpha to determine exactly which groups. First let's do a Bonferroni correction on a pairwise t-test.

```
> pairwise.t.test(vol_acid, qual, p.adj='bonferroni')

        Pairwise comparisons using t tests with pooled SD

data:  vol_acid and qual

  3       4       5       6       7
4 0.0118  -       -       -       -
5 7.8e-08 1.0e-05 -       -       -
6 3.5e-12 2.0e-15 < 2e-16 -       -
7 < 2e-16 < 2e-16 < 2e-16 5.2e-11 -
8 2.5e-11 2.9e-08 0.0014  0.8879  1.0000

P value adjustment method: bonferroni
```

It seems that every pairwise group has a low p-value suggesting a difference in population mean, except (6,8) and (7,8). Comparing this to a tukey correction,

```
> TukeyHSD(VolAnova)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = vol_acid ~ as.factor(qual))

$`as.factor(qual)`
           diff         lwr         upr     p adj
4-3 -0.19053774 -0.35217798 -0.02889749 0.0102247
5-3 -0.30745888 -0.45680111 -0.15811665 0.0000001
6-3 -0.38701567 -0.53643072 -0.23760063 0.0000000
7-3 -0.48058040 -0.63251748 -0.32864332 0.0000000
8-3 -0.46116667 -0.64607647 -0.27625687 0.0000000
5-4 -0.11692115 -0.18377920 -0.05006310 0.0000099
6-4 -0.19647794 -0.26349848 -0.12945740 0.0000000
7-4 -0.29004267 -0.36251178 -0.21757355 0.0000000
8-4 -0.27062893 -0.39852940 -0.14272846 0.0000000
6-5 -0.07955679 -0.10538865 -0.05372493 0.0000000
7-5 -0.17312152 -0.21090121 -0.13534183 0.0000000
8-5 -0.15370778 -0.26566341 -0.04175215 0.0013080
7-6 -0.09356473 -0.13163123 -0.05549822 0.0000000
8-6 -0.07415099 -0.18620374  0.03790175 0.4098254
8-7  0.01941374 -0.09598053  0.13480800 0.9968509
```

The same results that we saw in the Bonferroni correction appear here, but with the tukey correction we can observe the confidence interval for the differences among each pairs population averages. For 8-6 and 8-7 the confidence interval includes zero and the p-value is not smaller than 0.05. However, with the rest of the quality pairing the confidence interval lies below zero suggesting that the amount of volatile acidity in higher qualities will be lower than those of lower qualities. The biggest difference being between a red wine of quality 7 minus a red wine of quality 3. Therefor we have validated the assumption that too much acetic acid will cause an unpleasant taste in this specific red wine.

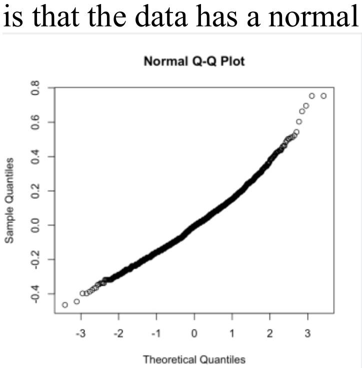
The next thing that must be done in order to confirm these results is to check assumptions made when performing an analysis of variance. The first assumption that we cannot check is that the data is random and independent. The next assumption is that there is equal variance in acidity among the different qualities of wine. Observing the standard deviations within the sample checks this assumption.

```
> # Checking for equal variance among each group
> sd(redWineQuality$volatile.acidity[redWineQuality$quality == 3])
[1] 0.3312556
> sd(redWineQuality$volatile.acidity[redWineQuality$quality == 4])
[1] 0.22011
> sd(redWineQuality$volatile.acidity[redWineQuality$quality == 5])
[1] 0.1648012
> sd(redWineQuality$volatile.acidity[redWineQuality$quality == 6])
[1] 0.1609623
> sd(redWineQuality$volatile.acidity[redWineQuality$quality == 7])
[1] 0.1452244
> sd(redWineQuality$volatile.acidity[redWineQuality$quality == 8])
[1] 0.1449138
```

For this sample, equal variance is skeptical but to account for this skepticism we can re-do the Bonferroni correction.

```
> pairwise.t.test(vol_acid, qual, p.adj='bonferroni', pool.sd = FALSE)

        Pairwise comparisons using t tests with non-pooled SD

data:  vol_acid and qual

  3      4       5       6       7
4 1.0000 -       -       -       -
5 0.2495 0.0056  -       -       -
6 0.0743 5.6e-07 < 2e-16 -       -
7 0.0193 5.7e-12 < 2e-16 1.6e-12 -
8 0.0231 5.9e-06 0.0048  0.7002  1.0000

P value adjustment method: bonferroni
>
```

Now it is noted that many other pairings have a p-value higher than 0.05 to show that maybe not having equal variances will affect this ANOVA. 3-4, 3-5, 4-5, 3-6, 6-8, and 7-8 are all pairs that we cannot say have a difference in the mean value of acetic acid. Moving on, the last assumption is that the data has a normal distribution. To check this, a Q-Q plot of the residuals will work.



Since this plot forms a somewhat straight line, we can assume normality and this concludes the check of assumptions for ANOVA.

Research Question #3.)

Using both variables, alcohol percentage and volatile acidity, is there a way to create an effective model for predicting the quality of *vinho verde* red wine? To create a model for predicting a categorical variable we must use a logistic regression model instead of a linear regression model. For this question we can even use both a binomial logistic regression model and a multinomial logistic regression model. The assumptions made for these models, there cannot be multicollinearity in between the two predictors we are using. To test this, we can check for correlation in between the two variables.

```
> cor(redWineQuality$volatile.acidity, redWineQuality$alcohol)
[1] -0.202288
```

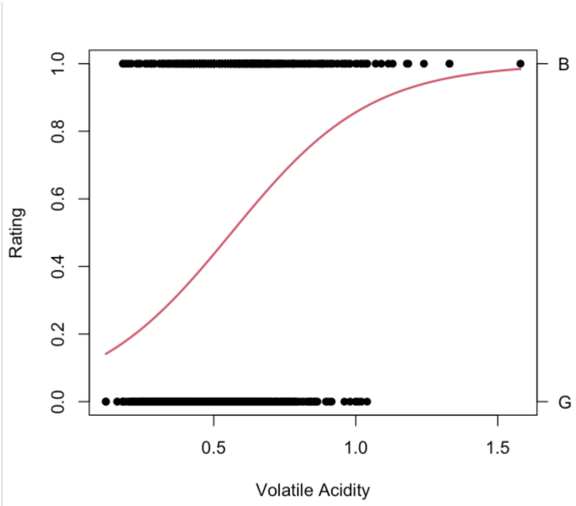
The correlation between volatile acidity and alcohol percentage is negative, but weak so it won't affect results.

Binomial Logistic Regression Models:

To first make a binomial logistic regression model, there can only be two groups that will fit into 1s and 0s. To do this I was able to subset the data where wines with quality greater that 5 will be assigned a 0 rating and wines with quality less than or equal to 5 will have a 1 rating. Creating the model that just uses volatile acidity as a predictor,

```
> logmodVA = glm(rating~ volatile.acidity, family=binomial(link=logit), data = redWineQuality)
> summary(logmodVA)

Call:
glm(formula = rating ~ volatile.acidity, family = binomial(link = logit),
    data = redWineQuality)

Coefficients:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)       -2.2936     0.1843  -12.45   <2e-16 ***
volatile.acidity   4.0769     0.3347   12.18   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2209.0  on 1598  degrees of freedom
Residual deviance: 2033.4  on 1597  degrees of freedom
AIC: 2037.4

Number of Fisher Scoring iterations: 4
```

Interpreting these results, we see that the p-value being small suggests that volatile acidity will be useful in predicting a "good" or "bad" wine. We can also graph the line from this model to visualize how it will predict the odds of a wine being good or bad.



The plot distributes values for volatile acidity from the sample if it has quality above or below 5. One thing that is noticeable in the plot is the outlier for the bad wine. There is one value that sticks out which seems to be influencing the model. For now, we can continue to interpret the model. The red line represents how the model is using the data to create a predictive model for quality. Interpreting the slope of this line which is seen in the summary of the model as 4.0769. The odds that a wine is "bad" increases by $e^{4.0769}$ when volatile acidity increases by 0.1 grams per liter.

```
> exp(coef(logmodVA)[2])
volatile.acidity
        58.96154
```

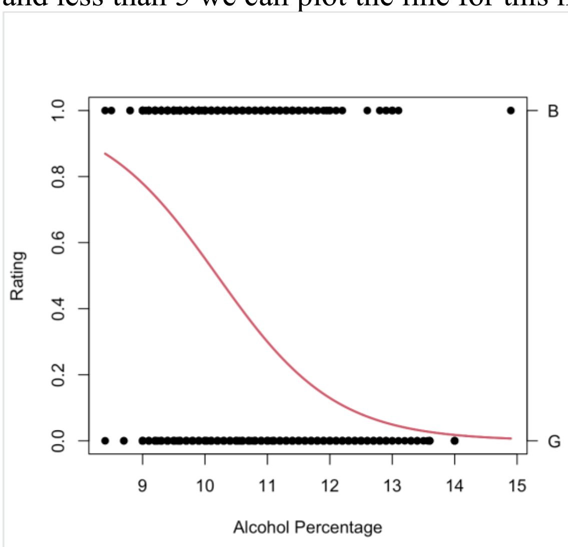
If volatile acidity is increased by .1, the odds that the wine is a bad are 58.96 times what they were before. The AIC of 2037 and Residual deviance of 2033.4 are both ways to rate the model created that also take size and interpretability as factors. For this model the AIC and deviance are high suggesting there are better model to predict "good" vs "bad" red wine. However, we can use this volatile acidity model to predict the probability a red wine is "bad" using the predict function.

```
> # average value
> predict(logmodVA, newdata=data.frame(volatile.acidity = 0.5), type = "response")
        1
0.4365528
> # smaller value
> predict(logmodVA, newdata=data.frame(volatile.acidity = .12), type = "response")
        1
0.1413186
> # higher value
> predict(logmodVA, newdata=data.frame(volatile.acidity = 1.5), type = "response")
        1
0.9785788
```

The result for each function is the probability of the red wine being a 1. For this example, being "bad". When acetic acid levels are low the wines have a low probability of being bad, but when the acetic acid levels are high, the wine has a high probability of being "bad". The next model that we can compare that to is a binomial logistic regression model to predict "good" vs "bad" using alcohol percentage as the predictor now.

```
> logmodALC = glm(rating~ alcohol, family=binomial(link=logit), data = redWineQuality)
> summary(logmodALC)

Call:
glm(formula = rating ~ alcohol, family = binomial(link = logit),
    data = redWineQuality)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) 10.76302    0.68326   15.75   <2e-16 ***
alcohol     -1.05559    0.06663  -15.84   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2209  on 1598  degrees of freedom
Residual deviance: 1865  on 1597  degrees of freedom
AIC: 1869

Number of Fisher Scoring iterations: 4
```

Interpreting some values/coefficients from this model we can see that the p-value is low once again suggesting that when adding alcohol percentage to the model it will have some effect on the quality of wine. Using a visualization of alcohol percentage given the wine has quality more and less than 5 we can plot the line for this model.



The line represents how the model predicts the probability that a wine is good or bad based on alcohol percentage. Interpreting the slope which from the summary is -1.056. The odds that a wine is "bad" increases by $e^{-1.056}$ when alcohol percentage increases by 1.

```
> exp(coef(logmodALC)[2])
  alcohol
0.3479873
```

If alcohol percentage is increased by 1, the odds that the wine is a bad are 0.3479 times what they were before. The residual deviance for this model is the same as the volatile acidity model, but the AIC of this model is smaller so this could suggest that alcohol percentage might be a better predictor of the quality of wine versus volatile acidity. Once again, we can use the predict function to find the probability that a wine is “bad” at different alcohol percentages.

```
> # average value
> predict(logmodALC, newdata=data.frame(alcohol = 10), type = "response")
       1
0.551598
> # smaller value
> predict(logmodALC, newdata=data.frame(alcohol = 8.5), type = "response")
        1
0.8569902
> # higher value
> predict(logmodALC, newdata=data.frame(alcohol = 14), type = "response")
         1
0.01771922
```

Based on these probabilities and the slope of the model line it can be said that when the alcohol percentage increases there is a smaller probability that the wine is “bad”. However, would a model using both predictors be a more efficient way to predict quality? We can make another binomial logistic regression model, but this time using volatile acidity and alcohol percentage as predictors.

```
> logmodBOTH = glm(rating~ volatile.acidity + alcohol, family=binomial(link=logit), data = redWineQuality)
> summary(logmodBOTH)

Call:
glm(formula = rating ~ volatile.acidity + alcohol, family = binomial(link = logit),
    data = redWineQuality)

Coefficients:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)       8.34395    0.72630  11.488   <2e-16 ***
volatile.acidity  3.54124    0.35470   9.984   <2e-16 ***
alcohol          -1.00468    0.06884 -14.594   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2209.0  on 1598  degrees of freedom
Residual deviance: 1752.5  on 1596  degrees of freedom
AIC: 1758.5

Number of Fisher Scoring iterations: 4
```

Based on the chi squared test in this model, the p-values suggest that when alcohol is added, even when volatile acidity is already in use, it will have significance on the prediction for wine quality. Along with that, the residual deviance of 1752.5 and AIC of 1758.5 are both smaller than they were before making this a “better” model. Using an ANOVA test we can see if adding another predictor is actually useful or not.

```
> anova(logmodALC, logmodBOTH, test='Chisq')
Analysis of Deviance Table

Model 1: rating ~ alcohol
Model 2: rating ~ volatile.acidity + alcohol
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1      1597     1865.0
2      1596     1752.5  1   112.53 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
> anova(logmodVA, logmodBOTH, test='Chisq')
Analysis of Deviance Table

Model 1: rating ~ volatile.acidity
Model 2: rating ~ volatile.acidity + alcohol
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1      1597     2033.4
2      1596     1752.5  1   280.87 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Comparing the model with 2 predictors to both original models we can see that the p-value is small. This means that adding a second predictor to the model with have significance in the model. So a model using both alcohol percentage and volatile acidity will be a better binomial model to predict the quality of red wine.

<u>Multinomial Logistic Regression Models:</u>

Given that this model only predicts “good” vs “bad”, what if we used a narrower approach to predict the actual quality of the red wine from 3 to 8? A multinomial logistic model will be useful given that we want to predict more than 2 groups. Using volatile acidity as the only predictor for quality the **1st** model looks like this.

```
> multiVA = multinom(quality ~ volatile.acidity , data = redWineQuality)
# weights:  18 (10 variable)
initial  value 2865.023391
iter  10 value 1827.124252
iter  20 value 1757.173521
iter  30 value 1756.399639
final  value 1756.387934
converged
> summary(multiVA)
Call:
multinom(formula = quality ~ volatile.acidity, data = redWineQuality)

Coefficients:
  (Intercept) volatile.acidity
4    4.105842        -3.129622
5    8.722341        -6.395566
6   10.239159        -9.347725
7   11.070658       -13.801515
8    8.234602       -12.754052

Std. Errors:
  (Intercept) volatile.acidity
4    1.199385         1.380829
5    1.158060         1.324324
6    1.168527         1.355705
7    1.189372         1.456675
8    1.390050         2.169071

Residual Deviance: 3512.776
AIC: 3532.776
```

Interpreting these results, the residual deviance and AIC are much higher than those for binomial logistic regression, but this is probably because the model is a more complicated and more specific that it was before. To find out how volatile acid effects the odds of each quality of wine compared to a wine of quality 3 we must take the coefficients given under "volatile.acidity" then substitute them for x in an equation like this,

$$e^x$$

Using r to calculate these values,

```
> exp(coef(multiVA)[,2])
           4            5            6            7            8
4.373434e-02 1.668942e-03 8.716347e-05 1.014094e-06 2.890583e-06
```

To interpret some values here, the odds of quality=8 vs quality-3 are multiplied by 2.89 x 10^-6 if volatile acidity increases by 0.1. The odds of quality=4 vs quality-3 are multiplied by 4.37 x 10^-2 if volatile acidity increases by 0.1. The same applies for all the other groups when we compare their odds to the base group, wine of quality 3. Next, we can predict the probability for each quality of wine given a value of volatile acidity.

```
> # average value
> round(predict(multiVA, data.frame(volatile.acidity = 0.5), type='probs'),3)
    3     4     5     6     7     8 
0.002 0.021 0.420 0.438 0.108 0.011 
> # smaller value
> round(predict(multiVA, data.frame(volatile.acidity = .12), type='probs'),3)
    3     4     5     6     7     8 
0.000 0.002 0.114 0.363 0.489 0.033 
> # bigger value
> round(predict(multiVA, data.frame(volatile.acidity = 1.5), type='probs'),3)
    3     4     5     6     7     8 
0.501 0.278 0.210 0.011 0.000 0.000 
>
```

This still backs the conclusion that when volatile acidity increases, the quality of wine decreases.

The **2nd** multinomial model will be using alcohol percentage as the indicator for the quality of red wine.

```
> multiALC = multinom(quality ~ alcohol, data = redWineQuality)
# weights:  18 (10 variable)
initial  value 2865.023391
iter  10 value 1690.904239
iter  20 value 1663.954061
iter  30 value 1663.425464
iter  40 value 1662.920673
iter  40 value 1662.920673
iter  40 value 1662.920673
final  value 1662.920673
converged
> summary(multiALC)
Call:
multinom(formula = quality ~ alcohol, data = redWineQuality)

Coefficients:
  (Intercept)    alcohol
4   -2.986676  0.4607999
5    5.255497 -0.1042167
6   -4.641628  0.8576551
7  -13.596426  1.5636982
8  -21.780886  2.0544449

Std. Errors:
  (Intercept)   alcohol
4    4.526909 0.4517613
5    4.272710 0.4280437
6    4.268257 0.4272560
7    4.335791 0.4323632
8    5.026612 0.4803601

Residual Deviance: 3325.841
AIC: 3345.841
```

For this model, the residual deviance and AIC are smaller than our model with volatile acidity once again showing alcohol might be a better predictor. To find out how alcohol effects the odds of each quality of wine compared to a wine of quality 3 we must take the coefficients given under “alcohol” then substitute them for x in an equation like this,

$$e^x$$

Using r to calculate these values,

```
> exp(coef(multiALC)[,2])
       4        5        6        7        8
1.585342 0.901030 2.357626 4.776453 7.802505
```

The odds of quality being 8 rather than 3 are multiplied by 7.80 if alcohol percentage increases by 1. The odds of quality being 4 rather than 3 are multiplied by 1.59 if alcohol percentage is increased by 1. However, the odds of a quality being 5 rather than 3 are multiplied by 0.90 if alcohol percentage is increased by 1. There is a slight bit of inconsistency in the trend of alcohol for predicting quality, but overall increased alcohol percentage seems to be in higher quality red wine.

Predicting the probabilities for each quality given alcohol percentage we see these results.

```
> # average value
> round(predict(multiALC, data.frame(alcohol = 10), type='probs'),3)
    3     4     5     6     7     8
0.008 0.038 0.509 0.385 0.058 0.002
> # smaller value
> round(predict(multiALC, data.frame(alcohol = 8.5), type='probs'),3)
    3     4     5     6     7     8
0.010 0.026 0.811 0.145 0.008 0.000
> # bigger value
> round(predict(multiALC, data.frame(alcohol = 14), type='probs'),3)
    3     4     5     6     7     8
0.000 0.005 0.007 0.235 0.594 0.160
```

The probabilities won't be exceedingly large for any value that won't extrapolate the data because this model cannot be that confident in the effectiveness of just one predictor. However, the results still show that when there is a higher alcohol percentage the wine seems to be of higher quality.

The **3rd** multinomial model will include both volatile acidity and alcohol percentage as predictors for the quality of wine.

```
> multiBOTH = multinom(quality ~ volatile.acidity + alcohol, data = redWineQuality)
# weights:  24 (15 variable)
initial  value 2865.023391
iter  10 value 1653.578004
iter  20 value 1566.603590
iter  30 value 1565.954610
iter  40 value 1565.936024
final  value 1565.933028
converged
> summary(multiBOTH)
Call:
multinom(formula = quality ~ volatile.acidity + alcohol, data = redWineQuality)

Coefficients:
  (Intercept) volatile.acidity   alcohol
4   -3.849317        -3.780530 0.8416283
5    6.012096        -6.937605 0.3131149
6   -2.065155        -9.610071 1.2379356
7   -8.884520       -13.399807 1.9038722
8  -17.677348       -12.292787 2.4075427

Std. Errors:
  (Intercept) volatile.acidity   alcohol
4    5.056448         1.488513 0.5381474
5    4.867728         1.438896 0.5198578
6    4.881725         1.468904 0.5212136
7    4.955384         1.573368 0.5263200
8    5.611036         2.259254 0.5664459

Residual Deviance: 3131.866
AIC: 3161.866
```

For this last model, the AIC and residual deviance suggest this model will be better than just using one predictor, but in the long run using only two predictors for the quality of wine won't be the best model. This makes sense because a lot of factors play into the quality of wine and just using two variables will not be the most accurate model, but for this case we can say that they might be more accurate than just one variable. We can now predict the probabilities of each quality using a value for alcohol percentage and volatile acidity.

```
> # average values
> round(predict(multiBOTH, data.frame(volatile.acidity = 0.5, alcohol = 10), type='probs'),3)
    3     4     5     6     7     8
0.002 0.025 0.497 0.421 0.054 0.002
> # "bad" values (high VA, low ALC)
> round(predict(multiBOTH, data.frame(volatile.acidity = 1.2, alcohol = 5), type='probs'),3)
    3     4     5     6     7     8
0.671 0.010 0.318 0.000 0.000 0.000
> # "good" values (low VA, high ALC)
> round(predict(multiBOTH, data.frame(volatile.acidity = 0.2, alcohol = 14), type='probs'),3)
    3     4     5     6     7     8
0.000 0.000 0.002 0.125 0.716 0.157
```

When the volatile acidity is low and alcohol percentage is high this will have the highest probability of being a higher quality of *vinho verde* red wine. In conclusion, when using an interpretable model with multinomial logistic regression, this model may not be the most accurate for quality as a whole, but it can be used to interpret the significance of both volatile acidity and alcohol percentage on the quality of wine. The best model for quality of wine most likely involves most of the variables explored in the beginning of the project, but that model would be complicated and very hard to interpret.

Works Cited

P. Cortez, A. Cerdeira, F. Almeida, T. Matos and J. Reis. Modeling wine preferences by data mining from physicochemical properties. In Decision Support Systems, Elsevier, 47(4):547-553, 2009.